



# Joel Oskarsson

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## Research Interests

In my research I work with probabilistic machine learning methods for modeling data with spatial and temporal dependencies. I am motivated by applications of machine learning to earth system modeling, such as weather forecasting and climate modeling. Part of my research is in core machine learning method development, where unique challenges from earth science act as an underlying motivation. These challenges include: irregular observations, high-dimensional data, complex temporal dynamics and the need for accurate uncertainty estimates. I also enjoy developing methods specifically for earth science applications, tailoring the machine learning methods using domain knowledge to solve relevant problems in new ways.

## Education

### 2020–2025 **Doctoral Studies in Computer Science (in progress)**

Linköping University, 240 ECTS

Linköping, Sweden

- PhD thesis: *Modeling Spatio-Temporal Systems with Graph-based Machine Learning*.
- At the Division of Statistics and Machine Learning (STIMA), Department of Computer and Information Science. Supervised by Fredrik Lindsten (main supervisor), Per Sidén and Tomas Landelius.
- Affiliated PhD Student in the Wallenberg AI, Autonomous Systems and Software Program (WASP).

### 2015–2020 **Master's program in Computer Science and Engineering**

Linköping University, 300 ECTS

Linköping, Sweden

- Master's thesis: *Probabilistic Regression using Conditional Generative Adversarial Networks*.

2018–2019 **Exchange Studies**

ETH Zürich

Zürich, Switzerland

First year of my master's as an exchange student at ETH.

## Employment

2020 – 2025 **PhD Student, Linköping University**

Linköping, Sweden

Research + 20% of employment teaching.

2016 – 2019 **Teaching Assistant, Linköping University**

Linköping, Sweden

Multiple periods.

2018, Summer **Summer Intern, Ericsson**

Linköping, Sweden

Internship at Ericsson Research, working with GNSS positioning.

## Research Visits

2024 (1 week) European Centre for Medium-Range Weather Forecasts (ECMWF)

Bonn, Germany

Hosted by Mariana Clare

2024 (1 week) ETH Zürich + MeteoSwiss

Zürich, Switzerland

Hosted by Simon Adamov

2023 (5 weeks) University College London (UCL)

London, U.K.

Hosted by Marc Deisenroth

2022 (1 week) Finnish Center for AI + Aalto University

Helsinki, Finland

WASP International Study Visit

## Publications

### *Journal papers*

- [1] Theodor Westny, **Joel Oskarsson**, Björn Olofsson, and Erik Frisk. “MTP-GO: Graph-Based Probabilistic Multi-Agent Trajectory Prediction with Neural ODEs”. In: *IEEE Transactions on Intelligent Vehicles* (2023).

*Peer-reviewed conference papers*

- [2] Martin Andrae, Tomas Landelius, **Joel Oskarsson**, and Fredrik Lindsten. “Continuous Ensemble Weather Forecasting with Diffusion models”. In: *International Conference on Learning Representations*. 2025.
- [3] **Joel Oskarsson**, Tomas Landelius, Marc Peter Deisenroth, and Fredrik Lindsten. “Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks”. In: *Advances in Neural Information Processing Systems*. Vol. 37. [Spotlight]. 2024.
- [4] **Joel Oskarsson**, Per Sidén, and Fredrik Lindsten. “Temporal Graph Neural Networks for Irregular Data”. In: *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*. 2023.
- [5] Theodor Westny, **Joel Oskarsson**, Björn Olofsson, and Erik Frisk. “Evaluation of Differentially Constrained Motion Models for Graph-Based Trajectory Prediction”. In: *2023 IEEE Intelligent Vehicles Symposium (IV)*. 2023.
- [6] **Joel Oskarsson**, Per Sidén, and Fredrik Lindsten. “Scalable Deep Gaussian Markov Random Fields for General Graphs”. In: *Proceedings of the 39th International Conference on Machine Learning*. PMLR, 2022.

*Peer-reviewed workshop papers*

- [7] Erik Larsson, **Joel Oskarsson**, Tomas Landelius, and Fredrik Lindsten. “Diffusion-LAM: Probabilistic Limited Area Weather Forecasting with Diffusion”. In: *ICLR 2025 Workshop on Tackling Climate Change with Machine Learning*. 2025.
- [8] Vignesh Gopakumar, **Joel Oskarsson**, Ander Gray, Lorenzo Zanisi, Stanislas Pamela, Daniel Giles, Matt Kusner, and Marc Deisenroth. “Valid Error Bars for Neural Weather Models using Conformal Prediction”. In: *ICML Workshop on Machine Learning for Earth System Modeling*. 2024.
- [9] **Joel Oskarsson**, Tomas Landelius, and Fredrik Lindsten. “Graph-based Neural Weather Prediction for Limited Area Modeling”. In: *NeurIPS 2023 Workshop on Tackling Climate Change with Machine Learning*. 2023.
- [10] **Joel Oskarsson**, Per Sidén, and Fredrik Lindsten. “Temporal Graph Neural Networks with Time-Continuous Latent States”. In: *ICML Workshop on Continuous Time Methods for Machine Learning*. 2022.

*Under review*

- [11] Simon Adamov\*, **Joel Oskarsson\***, Leif Denby, Tomas Landelius, Kasper Hintz, Simon Christiansen, Irene Schicker, Carlos Osuna, Fredrik Lindsten, Oliver Fuhrer, and Sebastian Schemm. “Building Machine Learning Limited Area Models: Kilometer-Scale Weather Forecasting in Realistic Settings”. In: *preprint* (2025).

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\*Equal contribution.

- [12] Vignesh Gopakumar, Ander Gray, **Joel Oskarsson**, Lorenzo Zanisi, Stanislas Pamela, Daniel Giles, Matt Kusner, and Marc Peter Deisenroth. “Uncertainty Quantification of Pre-Trained and Fine-Tuned Surrogate Models using Conformal Prediction”. In: *preprint* (2024).

## Invited talks

- 2025-05-02 *Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks*, Google DeepMind  
Online
- 2024-11-07 *Frontiers in Machine Learning for Weather Forecasting*, RISE Research Institutes of Sweden, Learning Machines Seminars  
Online
- 2024-08-22 *Graph-based Machine Learning for Weather*, European Centre for Medium-Range Weather Forecasts (ECMWF)  
Bonn, Germany
- 2024-05-02 *Graph-based Neural Weather Prediction for Limited Area Modeling*, MeteoSwiss  
Zürich, Switzerland
- 2024-04-30 *Graph-based Neural Weather Prediction for Limited Area Modeling*, ETH Zürich  
Zürich, Switzerland
- 2024-04-17 *Graph-based Machine Learning for Spatio-Temporal Data, with Application to Traffic and Weather Forecasting*, University of New South Wales (UNSW)  
Sydney, Australia (Online)
- 2024-03-05 *Graph-based Neural Weather Prediction for Limited Area Modeling*, Uppsala University  
Uppsala, Sweden
- 2024-01-11 *Neural Weather Prediction for Limited Area Modeling*, IEA Wind Task 51 Webinar: Forecasting for the Weather Driven Energy System  
Online
- 2023-10-10 *Graph-based Neural Weather Prediction for Limited Area Modeling*, Danish Meteorological Institute  
Copenhagen, Denmark

## Participation in Scientific Events

- 2025 *Royal Swedish Academy Of Sciences Symposium on AI for Science*  
Stockholm, Sweden  
Oral presentation

- 2025 *Workshop on Machine Learning for Earth System Modelling*  
Bonn, Germany  
Oral presentation
- 2025 *Learning on Graphs and Geometry Meetup*  
Uppsala, Sweden  
Oral presentation
- 2024 *Conference on Neural Information Processing Systems (NeurIPS)*  
Vancouver, Canada  
Poster presentation
- 2024 *ELLIIT Focus Period Symposium: Machine Learning for Climate Science*  
Linköping, Sweden  
Poster presentation
- 2024 *Large-Scale Deep Learning for the Earth System Workshop*  
Bonn, Germany  
Oral presentation
- 2024 *International Conference on Machine Learning (ICML)*  
*Machine Learning for Earth System Modeling Workshop at ICML 2024*  
Vienna, Austria  
Workshop poster presentation
- 2024 *ESA-ECMWF workshop on Machine Learning for Earth System Observation and Prediction*  
Frascati, Italy  
Oral presentation + poster presentation
- 2024 *ELLIIT Annual Workshop*  
Lund, Sweden  
Poster presentation
- 2024 *WASP Winter Conference*  
Norrköping, Sweden  
Poster presentation
- 2023 *Conference on Neural Information Processing Systems (NeurIPS)*  
*Tackling Climate Change with Machine Learning Workshop at NeurIPS 2023*  
Online attendance  
Workshop poster presentation
- 2023 *Large-Scale Deep Learning for the Earth System Workshop*  
Bonn, Germany  
Oral presentation
- 2023 *Nordic Conference in Mathematical Statistics (NORDSTAT)*  
Gothenburg, Sweden  
Oral presentation

- 2023 *International Conference on Artificial Intelligence and Statistics (AISTATS)*  
Valencia, Spain  
Poster presentation
- 2023 *Swedish Symposium on Image Analysis and Deep Learning*  
Norrköping, Sweden  
Poster presentation
- 2023 *WASP Winter Conference*  
Norrköping, Sweden  
Poster presentation
- 2022 *AI Day Finland*  
Helsinki, Finland
- 2022 *Learning on Graphs Conference (LoG)*  
Online attendance
- 2022 *International Conference on Machine Learning (ICML)*  
*Continuous Time Methods for Machine Learning Workshop at ICML 2022*  
Baltimore, U.S.  
Poster presentation + Workshop poster presentation (*Cancelled due to illness*)
- 2022 *Nordic Probabilistic AI Summer School*  
Helsinki, Finland  
Poster presentation

## Career Development Grants

- 2023 Research Visit Scholarship ( $\approx 60\,000$  SEK)  
Wallenberg AI, Autonomous Systems and Software Program (WASP)

## Other Academic Activities

- 2023– Co-organizer of monthly online meetups for researchers working on machine learning limited area weather forecasting models.
- 2023– Creator and maintainer of the *Neural-LAM* open source python library for machine learning weather forecasting models.
- 2022–2025 Organizer of the *STIMA Machine Learning Reading Group* at Linköping University.

## Reviewing

- 2025 Geophysical Research Letters
- 2025 International Conference on Learning Representations (ICLR)

- 2024 Conference on Neural Information Processing Systems (NeurIPS)
- 2023,2024 International Conference on Artificial Intelligence and Statistics (AISTATS)
- 2024,2025 Tackling Climate Change with Machine Learning Workshop (Climate Change AI)

## Media Coverage of Research

- 2024 *De vill göra väderprognoser mer pålitliga – med hjälp av AI*  
News coverage, SVT (Swedish Public Service Television)
- 2023 *Förbättrade väderprognoser med AI*  
News posting, Linköping University

## Scientific Outreach and Popular Science

- 2025 *Forskningsframsteg inom Maskininlärning för Väderprognoser*  
Popular science article about machine learning weather forecasting published in Polar-front, the journal of the Swedish meteorological society.
- 2025 *Doing a PhD in Machine Learning and AI within WASP*  
Talk for undergraduate students about machine learning research and doing a PhD within the AI area.

## Specific Skills and Knowledge

- Extensive knowledge of modern **machine learning** methods. Specific expertise in:
  - Probabilistic deep learning
  - Spatio-temporal models of physical systems
  - Bayesian models and inference methods
  - Graph neural networks
- Sound knowledge of good **software engineering** practices, acquired from my undergraduate studies and research experience.
- Programming languages and frameworks:  
*Python, PyTorch, SciPy, NumPy, PyTorch Lightning, Xarray, PyTorch Geometric, scikit-learn, Bash, R, C++, JavaScript*
- Highly accustomed to **Linux** environments.
- Used to working with **high-performance computing** systems.
- Languages:

- Swedish: Native
- English: Fluent
- German: Basic skills

## References

- *Prof. Fredrik Lindsten*

Position: Senior Associate Professor at Linköping University, Sweden

Relation: Main supervisor of my PhD

Webpage: <https://lindsten.netlify.app/>

E-mail: [fredrik.lindsten@liu.se](mailto:fredrik.lindsten@liu.se)

- *Prof. Marc Deisenroth*

Position: Google DeepMind Chair of Machine Learning and Artificial Intelligence at University College London, U.K.

Relation: Collaborator

Webpage: <https://www.deisenroth.cc/>

E-mail: [m.deisenroth@ucl.ac.uk](mailto:m.deisenroth@ucl.ac.uk)

- *Dr. Tomas Landelius*

Position: Researcher at the Swedish Meteorological and Hydrological Institute

Relation: Co-supervisor of my PhD

Webpage: <https://www.smhi.se/en/research/research-departments/meteorology/tomas-landelius-1.4817>

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